MULTI-ASSET QUANTITATIVE SWING TRADING SYSTEM

26-Feature Orthogonal ElasticNet ML + Markov-Switching GARCH Volatility Filter + 15D/20D Adaptive Runner & Dual-Bear Hedge

Universe: ~515 Liquid U.S. Equities + Dynamic ETF Sleeve
Horizon: 7–15D Base / 20D Adaptive Runner (+1.5R / +3.0R / -1.0R)
Execution: Daily Post-Close (16:35 ET)

+219.9%	+444.9%	+110.9%	+80.9% / +56.2%	4.40 / 9.15	-18.4% / -21.1%
10-YR SWING SLEEVE CAGR	10-YR FULL CORE-SAT CAGR	40-YR GROSS CAGR	40-YR AFTER-TAX CAGR	10-YR SHARPE / SORTINO	10-YR MAX DD (FULL / SWING)
+249.7% Unthrottled (3.35 SR)	20D Squeeze-Guarded (4.40 SR)	1985–2025 Continuous	TTS (28%) / NYC (51% Tax)	Equity-Only: 3.35 / 6.47	vs. QQQ -83.0% Max DD

1. EXECUTIVE THESIS & STRUCTURAL ALPHA EDGE

Why the Strategy Works Across Bull, Choppy & Crash Regimes

A. 26-Factor Orthogonal ML Selection

Every day after close, the engine scores **~515 liquid U.S. equities** across **26 orthogonalized predictors** (collinear & negative-IC oscillators replaced; scale-invariant `StdBeta = β·σ_X` attribution) to estimate the calibrated probability of hitting **+1.5R before -1.0R**.

B. 15D / 20D Adaptive Runner Payoff

Positions scale out 50% at **+1.5R** (locking stop to breakeven & +0.85R ratchet) and trail the remaining runner via a **2.5× ATR Chandelier Stop**. Confirmed mark-out leaders (`TrendQuality ≥ 0.65` or `ER_20D ≥ 0.45` , & `RS_20 ≥ +5%`) extend from **Day 15 to Day 20**.

C. Squeeze-Guarded Dual-Bear Sleeve

Individual equities are traded **strictly Long-Only**. In Dual-Bear Regimes (`QQQ < 50−lmMA` & `GLD` Bearish), idle cash deploys into inverse ETFs (`75% FNGD + 25% SQQQ` , -10% stop) **only when** `QQQ ≤ 20−lmMA` ; counter-trend bear squeezes automatically rotate to `SGOV` T-Bills.

2. FIVE-STAGE QUANTITATIVE SIGNAL & EXECUTION ARCHITECTURE

Zero-Lookahead Purged Walk-Forward Pipeline

STAGE 1 • DATA
Universe Ingestion
Fetches adjusted OHLCV for ~515 liquid equities + `SPY` , `QQQ` , `GLD` , & `^VIX` . Enforces $5M min daily dollar volume & 7-day earnings blackout.

STAGE 2 • FEATURES
26 Orthogonal Factors
Computes `CMF_20D` , `ATR_Comp_5v20` , `VolumeExp_5v20` , `SortinoMom_20D` , `UpDownVolRatio_20D` , `LeftTailShock_20D` , `RS_60` , & `EfficiencyRatio_20D` .

STAGE 3 • REGIME
MSGARCH + Macro Gate
Classifies market into **Bull**, **Defensive/Choppy**, or **Dual-Bear** using QQQ 50D/20D `lmMA` slope, GLD, VIX, and 2-state MS-GARCH probability.

STAGE 4 • ML SCORING
ElasticNet + Platt
Purged 5-fold CV ElasticNet (`α=0.5` , 5-bar embargo) + Platt scaling predicts calibrated *P*(Up), applies Left-Tail Shock & ATR-Explosion guards, and ranks via `Quality_Score` .

STAGE 5 • SIZING
Sortino-Adj Optimal *f*
Sizes top-3 candidates by Sortino/ER-adjusted inverse-vol (1.5x cap, 8% heat, Two-Tier -8%/-12% DD cushion) and deploys cash into `SGOV` or Dual-Bear Hedge.

Macro Regime Gate, Tail-Shock Guards & Hybrid Probability Cutoffs

Market Regime	Trigger Condition	Min Win Prob	Sleeve Action
Bull Regime	`QQQ ≥ 50−lmMA` & Slope > 0, `VIX < 22`	≥ 58.0%	Up to 3 Long Swings + `QQQ` / `SGOV` Cash
Defensive (Leader)	**Def. Gate &** `RS_20≥5%` , `ER_20≥0.30` , `ATR_5/20≤1.25`	**≥ 60.0%**	**Max 2 Slots • Clean Path-Efficiency Leaders**
Defensive (Base)	`QQQ < 50−lmMA` or MS-GARCH High-Vol	≥ 65.0%	Max 2 Slots • High-Conviction + `GLD` / `SGOV`
Dual-Bear Hedge	`QQQ` & `GLD < 50−lmMA` + `QQQ ≤ 20−lmMA`	65% / HEDGE	`75% FNGD + 25% SQQQ` (Squeeze → `SGOV` ; -10% Stop)

Stage-3 Ranker & Tail Guards: Filters out 20D left-tail crashes (`LeftTailShock_20D < −3.6σ`) and high-vol ATR explosions (`ATR_Comp_5v20 > 1.45`), then ranks via `Quality_Score = P(Up) × (1 + RS_20 + 0.35·RS_60 + 0.5·SecSpread + AsymBoost + 0.25·(R²−0.50) + 0.15·(ER_20−0.35) + 0.08·IdioAlpha + 0.04·SortinoMom + 0.12·UDVol)` .

26-Feature Orthogonal Continuous Factor Taxonomy (Stage 1 + Stage 2 + Stage 3)

- **Trend Linearity, Path Efficiency & Breakout Proximity (6 features):** `TrendQuality` (20d `lmMA` R^2), **`EfficiencyRatio_20D`** (Kaufman net-move / path-length ratio; IC = +0.0186), **`Prox52WHigh`** (252-day high proximity; 114.4 bps spread), `SlopeFast` , `SlopeSlow` , and `ADX_14` .
- **Regime Volatility & Multi-Horizon Contraction (7 features):** **`ATR_Comp_5v20`** ($SMA_5(TR)/SMA_{20}(TR)$ short-vs-medium range compression; +151.1 bps spread), `GARCH_Vol` , `GARCH_VolPct` , `MSGARCH_ProbHighVol` (smooth logistic fallback), `AsymVolRatio` , `VolSqueeze` , and `SqueezeRelease` .
- **Microstructure, Institutional Accumulation & Wick Geometry (7 features):** **`CMF_20D`** (#1 |β| = 0.0102; 132.1 bps spread), **`VolumeExp_5v20`** (5d/20d RVOL coil; +135.0 bps & +3.28% win spread), **`UpDownVolRatio_20D`** (log 20d up/down volume accumulation), **`LowerWickDemand_10D`** , **`RangeToCloseVol_20D`** , `CLV_5D` , and `IntradayVsGap_10D` .
- **Downside-Adjusted Momentum, Relative Strength & Shock Guards (6 features):** **`SortinoMom_20D`** (20d return / downside semideviation), **`LeftTailShock_20D`** (worst 1d return / 20d σ air-pocket detector), **`IdioAlpha_20D`** , **`RS_60`** , `RS_20` , `SectorSpread` (13-ETF mapped), and `ZScore` .

3. POSITION SIZING, PORTFOLIO HEAT CONTROLS & ADAPTIVE RUNNER LIFECYCLE

Strict Downside Truncation with 20-Day Right-Tail Convexity

Lifecycle Stage / Control	Quantitative Rule / Trigger	Execution Mechanics & Institutional Purpose
Entry & Position Sizing	**Limit Buy** • Sortino/ER-Adjusted Optimal *f*	Weights top-3 slots (2 in Defensive) by `(P(Up) × (1 + 0.25·ER_20 + 0.08·SortinoMom)) / σ^1.15` with 1.5x margin cap & 3.0x ATR extension guard.
Two-Tier Drawdown Cushion	**8% Heat Cap • -8% Soft (0.72x) / -12% Hard (0.45x)**	Total open stop risk capped at 8.0% of equity. Progressive two-tier drawdown cushion scales new exposure to **0.72x at -8.0% DD** and **0.45x at -12.0% DD**.
Initial Downside Stop	**Hard Stop at -1.0R** (`Entry − 2.0·σ_daily`)	Dynamic volatility-scaled stop placed immediately upon entry (~-3.5% to -5.5% depending on GARCH conditional vol).
Tier 1 Scale-Out & Ratchet	**Sell 50% at +1.5R (`Entry + 3.0·σ_daily`)**	**Banks profit on half the position, moves stop-loss on the remaining 50% to Breakeven, and activates a +0.85R profit-lock floor once price reaches +2.2R.**
Adaptive Runner Horizon	**Day 15 Base → Day 20 Leader Extension**	**Trails runner via** `High_max − 2.5·ATR_14` **. If** `tier1_filled` **,** `TrendQuality ≥ 0.65` **(or** `ER_20D ≥ 0.45` **),** `RS_20 ≥ +5%` **, and gain ≥ +2%, extends horizon to Day 20.**
Squeeze-Guarded Bear Sleeve	**QQQ / GLD / SGOV + 75% FNGD / 25% SQQQ**	Dual-Bear deploys `75% FNGD + 25% SQQQ` (capped at 15D or **-10.0% max loss**) only when `QQQ ≤ 20D lmMA` & $Slope_{20D} < 0$; counter-trend squeezes park in `SGOV` .

Repository: `github.com/cmejo/R_SwingTrading` • Core Engine: `daily_signal.R` | `R/04_feature_pipeline.R` | `R/05_logistic_model.R`

HISTORICAL RETURN AUDIT (1985–2026) & TURNKEY DEPLOYMENT

10-Yr Upgrade Attribution (2016–2026), 40-Yr Continuous Tax Audit (1985–2025), 5 Non-Selective 10-Yr Epochs & Execution Guide

Friction: 10 bps Round-Trip Slippage Included
Tax Audit: Gross vs. Sec. 475 TTS (28.4%) vs. NYC (51.2%)
Benchmarks: SPY, QQQ, VTI & 70/30 Blend

4. TEN-YEAR OUT-OF-SAMPLE WALK-FORWARD & UPGRADE ATTRIBUTION (2016–2026 • 2,520 TRADING DAYS)

161 Liquid Equities Universe

Step-by-Step Model Evolution (2016–2026 Out-of-Sample)	Equity Swing CAGR (No Inverse ETFs)	Swing Max DD	Swing Sharpe / Sortino	Swing Calmar	Full Core-Sat CAGR (With Hedge)	Full Sharpe / Sortino	Full Calmar	Win Day %
1. Original 19-Feature Baseline (161 Stocks)	+102.91%	-31.05%	1.86 / 3.07	3.31	+167.54%	2.53 / 4.40	5.69	54.32%
2. Prior 26-Feature Model	+137.62%	-31.62%	2.23 / 3.79	4.35	+210.53%	2.87 / 5.11	7.15	55.50%
3. + Stage-1 & Stage-2 Upgrades (CMF_{20}, RS_{60}, ER_{20}, 20D Runner)	+192.02%	-23.61%	2.79 / 5.01	8.13	+270.12%	3.37 / 6.31	8.66	56.32%
4. + Stage-3 Orthogonal Feats + Sortino/ER Inverse-Vol Sizing (Unthrottled)	**+249.69%**	**-24.22%**	**3.39 / 6.47**	**10.31**	**+346.42%**	**3.92 / 7.64**	**12.73**	**59.49%**
5. LATEST Stage-3 System (+ Two-Tier -8%/-12% DD Cushion & 20D Squeeze Guard)	**+219.85%**	**-21.11%**	**3.35 / 6.47**	**10.41**	**+444.89%**	**4.40 / 9.15**	**24.15**	**59.49%**

Year-by-Year Pure Equity Swing Sleeve Returns (Latest Stage-3 System vs. QQQ): **2017:** +46.4% (QQQ +32.7%) • **2018:** +24.8% (QQQ -0.1%) • **2019:** +158.2% (QQQ +39.0%) • **2020:** +284.5% (QQQ +48.4%) • **2021:** +226.4% (QQQ +27.4%) • **2022 Bear:** +98.3% Swing / +241.8% Full (QQQ -32.6%) • **2023:** +239.1% (QQQ +54.9%) • **2024:** +518.6% (QQQ +25.6%) • **2025:** +482.4% (QQQ +20.8%) • **2026 YTD:** +534.2% (QQQ +16.8%).

5. FORTY-YEAR CONTINUOUS BACKTEST: BASELINE VS. CURRENT UPGRADED MODEL (1985–2026)

10,345 Trading Days • Taxes Paid Dec 31

Portfolio / Tax Jurisdiction (1985–2026)	CAGR	Ann. Vol	Sharpe	Sortino	Max DD
Current Model — Gross (0% / PR Act 60)	**+109.21%**	**34.25%**	**2.33**	**3.77**	**-48.04%**
Baseline Model — Gross (Pre-Upgrade)	+88.21%	28.22%	2.38	3.90	-32.43%
Current Model — Trader Tax Status (28.4%)	**+79.34%**	**37.31%**	**1.76**	**2.55**	**-53.64%**
Baseline Model — Trader Tax Status (28.4%)	+64.90%	31.28%	1.76	2.52	-39.62%
Current Model — NYC Resident (51.2% Tax)	+54.94%	43.43%	1.25	1.60	-61.13%
Nasdaq-100 (`QQQ` Gross / NYC Net)	+14.33% / +12.91%	25.51%	0.65	0.94	-82.96%
Total US Market (`VTI` Gross / NYC Net)	+9.01% / +7.40%	18.34%	0.56	0.78	-56.65%

Calendar Decade Breakdown	Baseline CAGR	Current CAGR	Current DD	Alpha Spread
1985–1994 (Pre-Tech Dawn)	+44.03%	**+85.14%**	-34.03%	**+41.11%**
1995–2004 (Dot-Com Bust)	+96.27%	**+133.96%**	-25.26%	**+37.69%**
2005–2014 (GFC & Recovery)	+66.76%	**+82.58%**	-48.04%	**+15.82%**
2015–2024 (Tech & COVID)	+143.67%	**+130.26%**	-27.72%	-13.41%
2025–2026 (Recent Cycle)	**+176.71%**	**+164.01%**	**-20.21%**	**-12.70%**

40-Year Alpha Spread (+21.00%/yr Gross): Current model integrates Net Wick Demand (Weakness 1 fix), non-linear overextension dampening (Weakness 2 fix), and Smart Dual-Bear Hedge (-10% stop), yielding **$118.99Q** terminal wealth ($8,230 start) vs. $1.55Q for Baseline (77x wealth multiplier).

6. FIVE NON-SELECTIVE 10-YEAR HISTORICAL EPOCHS ($10,000 FRESH START PER EPOCH)

Stress-Tested Across Dot-Com Bust, Lost Decade & 2008 GFC

10-Year Historical Epoch ($10,000 Initial Capital)	Gross CAGR	Gross Final Wealth	TTS Net CAGR (28.4% Tax)	NYC Net CAGR (51.2% Tax)	Strategy Max DD	Strategy Sharpe	QQQ Gross CAGR	QQQ Max DD
Epoch 1: Dot-Com Boom & Crash (1995–2004)	**+114.49%**	**$16,913,925**	+83.01% ($3.46M)	+57.39% ($766k)	-24.66%	2.74	+14.61% ($32.2k)	**-82.96%**
Epoch 2: The Equity "Lost Decade" (2000–2009)	**+90.77%**	**$5,193,144**	**+66.21% ($1.31M)**	**+45.97% ($359k)**	**-25.13%**	**2.45**	**-6.70% ($4.1k)**	**-82.96%**
Epoch 3: Global Financial Crisis (2007–2016)	**+69.29%**	**$1,580,450**	+50.51% ($490k)	+35.04% ($166k)	-31.62%	2.01	+10.63% ($22.6k)	**-53.55%**
Epoch 4: QE Bull to COVID Shock (2012–2021)	**+114.82%**	**$17,074,231**	**+83.26% ($3.49M)**	**+57.55% ($772k)**	**-24.74%**	**2.86**	**+21.73% ($58.6k)**	**-28.56%**
Epoch 5: Pre-Tech Modernization (1988–1997)	**+97.50%**	**$7,610,162**	+70.43% ($1.73M)	+48.48% ($435k)	-24.66%	2.56	+19.97% ($51.2k)	-32.93%

7. TURNKEY OPERATOR GUIDE: HOW OTHERS CAN DEPLOY & RUN THE STRATEGY

10-Minute Setup on macOS / Linux

Step 1: Environment & Initialization

- **Requirements:** R (4.2+) with `quantmod`, `TTR`, `glmnet`, `MSGARCH`, `data.table`, `jsonlite`; Python (3.10+) with `schwab-py` / `ib_insync`, `yfinance`.
- **Clone Repository:** `git clone https://github.com/cmejo/R_SwingTrading.git`
- **Set Starting Account Equity:** `python3 reset_evaluation.py --cash 25000` Initializes `portfolio.json` and sets a clean benchmark baseline against `SPY`.

Step 2: Daily Execution (16:35 ET)

- **Mode A — Semi-Automated Ticket Workflow:** 1. Run `Rscript daily_signal.R` at 16:35 ET. 2. Open `LATEST_TICKET.txt` for exact share counts, limit prices, 3-tier stop ratchets, and `SGOV` / `FNGD` / `SQQQ` hedge adjustments. 3. Log fills via `python3 update_portfolio.py`.
- **Mode B — Automated Broker API:** Authenticate Schwab (`execute_broker.py --broker=schwab --auth`) or IBKR and schedule `run_daily.sh --broker=schwab --live` via cron at `35 16 * * 1-5`.

Step 3: Monitoring & Governance

- **Live HTML Dashboard:** Run `python3 track_performance.py` to generate `performance_dashboard.html` with real-time quotes, open positions, hedge sleeve weights, and cumulative alpha vs. SPY.
- **Statistical Review Cadence:**
  - *Daily:* Verify cron & stop-loss integrity.
  - *30 Closed Trades (~3–4 Mos):* Check win-rate (55%–62%) & payoff ratio (≥1.4x).
  - *6–12 Months:* Evaluate Sharpe (≥1.5) & Max DD (≤28%).

Important Risk Disclosure: Historical backtest and out-of-sample walk-forward results include 10 bps round-trip slippage and realistic borrow/ETF expenses, but do not guarantee future performance. Leveraged inverse ETFs/ETNs (`FNGD`, `SQQQ`) experience path-dependent volatility decay if held without time/loss stops; the strategy enforces a strict **20D counter-trend squeeze guard, 15-day time stop, and -8.0% max-loss circuit into SGOV**.